

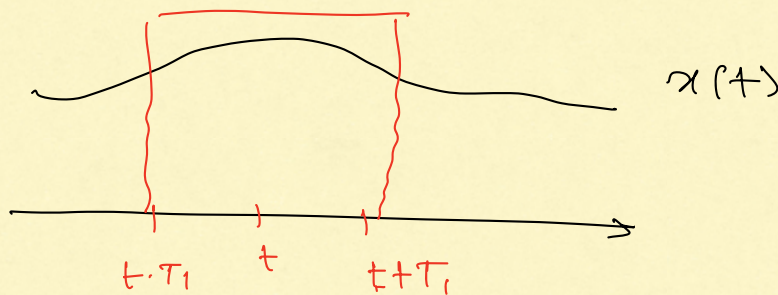
## I. Example Scenario

Suppose we have a noisy observation of a signal of interest.

$$y(t) = x(t) + n(t)$$

$\downarrow$  observation       $\downarrow$  signal of interest       $\rightarrow$  noise

- ① Assume  $x(t)$  varies slowly, say in a window of time:  $[t-T_1, t+T_1]$ ,  $x(t)$  doesn't vary significantly.



$\Rightarrow$  The average value of  $x$  in this window:

$$\frac{1}{2T_1} \int_{t-T_1}^{t+T_1} x(\tau) d\tau \approx x(t).$$

- ② Suppose the noise  $n(t)$  varies more rapidly and is equally spaced above & below the horizontal axis:  $t-T_1$

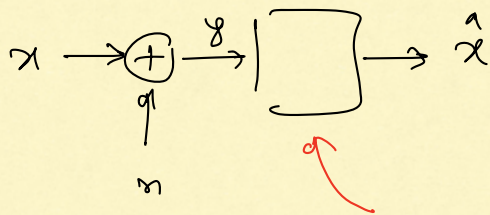
$$\frac{1}{2T_1} \int_{t-T_1}^{t+T_1} n(\tau) d\tau \approx 0.$$

$\Rightarrow$  For any choice of  $t$  we have:

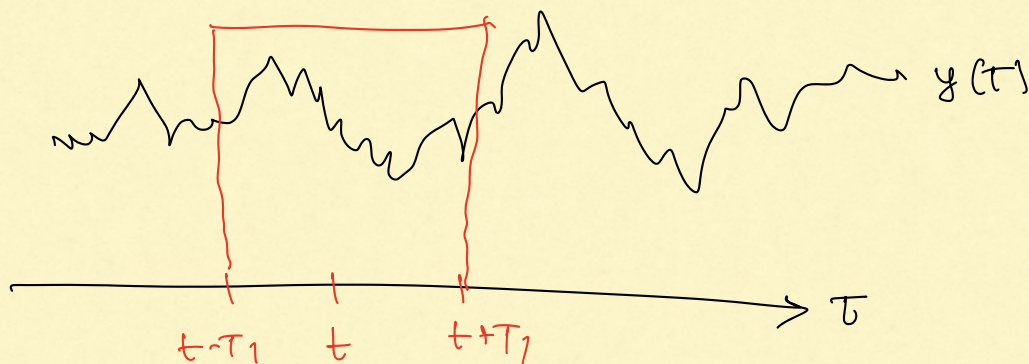
$$\frac{1}{2T_1} \int_{t-T_1}^{t+T_1} y(t) dt \approx x(t).$$

$\downarrow$

$\therefore$  this can be used an approximation to or estimate of  $x(t)$ .



This system is called the moving average filter.

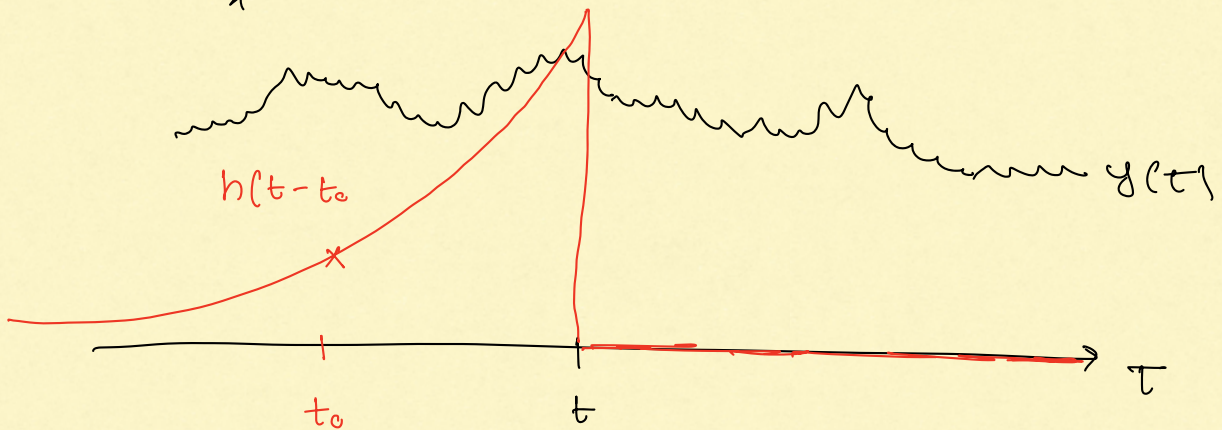


Suppose we want to make two improvements to this system:

- ① Estimate  $x$  at time  $t$  using only current and past values of  $y$ . In particular, do not use future values of  $y$ .
- ② If  $x$  is varying slowly enough, even the values of  $y$  beyond  $t-T_1$  might still



Carry some information about  $x$ . Let us use all previous values of  $y$  to estimate  $x$ .

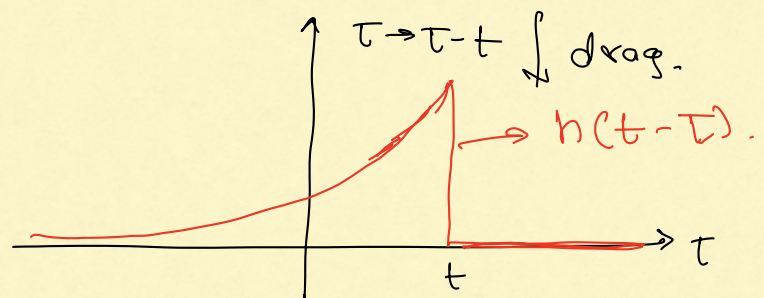
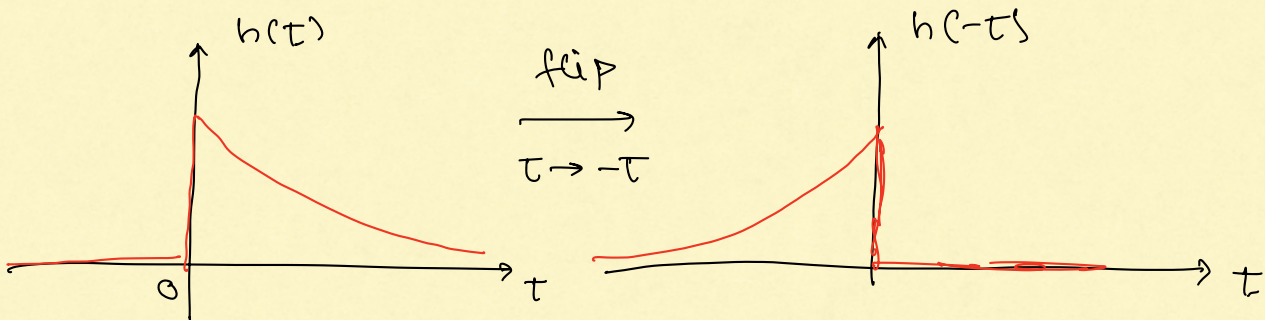


Let  $h$  denote the weightage being given to the observations:

$$\begin{aligned} \text{if } t_0 > t: h(t-t_0) &= 0 \\ \text{if } t_0 \leq t: h(t-t_0) &= e^{-(t-t_0)} \end{aligned}$$

Define the function  $h$  as below:

$$h(\tau) = \begin{cases} 0, & \text{if } \tau < 0 \\ e^{-\tau}, & \text{if } \tau \geq 0. \end{cases}$$



Flipped & dragged signal:

$h(-t)$  with  $t$  replaced by  $t - t$

$$\Rightarrow h(-(t-t)) = h(t-t)$$

Therefore, we have:

$$\hat{x}(t) = \int_{\tau=-\infty}^{+\infty} y(\tau) \cdot h(t-\tau) d\tau.$$

↓  
This is called convolution between  $y$  &  $h$  and is denoted as  $y * h$ .

## II. Convolution

Defn: Given two signals  $y$  and  $h$  their convolution is a signal denoted as  $y * h$  and is defined as:

$$(y * h)(t) = \int_{\tau=-\infty}^{+\infty} y(\tau) h(t-\tau) d\tau.$$

We immediately see the following:

$$\text{Use } u = t - \tau$$

$$du = -d\tau$$

$$\therefore (y * h)(t) = \int_{u=-\infty}^{+\infty} y(t-u) h(u) (-du)$$



$$= \int_{u=-\infty}^{+\infty} h(u) \cdot y(t-u) du$$

$$= h * y$$

$\Rightarrow$  Commutativity holds!

$$\therefore y * h = h * y.$$

$$\therefore (y * h)(t) = \int_{\tau=-\infty}^{+\infty} y(\tau) h(t-\tau) d\tau$$

$$= \int_{\tau=-\infty}^{+\infty} h(\tau) \cdot y(t-\tau) d\tau$$

Also, convolution is linear in its arguments:

Consider  $x * (\alpha_1 h_1 + \alpha_2 h_2)$  where  $x, h_1, h_2$  are signals and  $\alpha_1, \alpha_2 \in \mathbb{C}$  are scalars.

$$\left[ x * (\alpha_1 h_1 + \alpha_2 h_2) \right](t) = \int_{\tau=-\infty}^{+\infty} x(\tau) \left[ \alpha_1 h_1(t-\tau) + \alpha_2 h_2(t-\tau) \right] d\tau$$

$$= \alpha_1 (x * h_1)(t) + \alpha_2 (x * h_2)(t).$$

$$\Rightarrow x * (\alpha_1 h_1 + \alpha_2 h_2) = \alpha_1 x * h_1 + \alpha_2 x * h_2.$$

Similarly:  $(\alpha_1 x_1 + \alpha_2 x_2) * h = \alpha_1 x_1 * h + \alpha_2 x_2 * h.$

Observe the similarity between multiplication & convolution:

| Convolution                             | Multiplication                                    |
|-----------------------------------------|---------------------------------------------------|
| $y(t) = (x * h)(t)$                     | $y(t) = x(t) \cdot h(t)$                          |
| $x * h = h * x$                         | $x \cdot h = h \cdot x$                           |
| $(x_1 + x_2) * h = x_1 * h + x_2 * h$   | $(x_1 + x_2) \cdot h = x_1 \cdot h + x_2 \cdot h$ |
| $(\alpha x) * h = \alpha \cdot (x * h)$ | $(\alpha x) \cdot h = \alpha \cdot (x \cdot h)$   |

This similarity is not accidental. In future lectures we will learn Fourier Transform and observe that convolution in time domain becomes multiplication in frequency domain, and multiplication in time domain becomes convolution in frequency domain.